

The Team

Creative Director: He is the ideas man who lends his views visualises and conceptualises the whole project. He supervises the entire project and guides his team accordingly.

Graphic designer: This is the person responsible for the visual look and feel of the multimedia presentation. A good sense of visual appeal, colours and design harmony, typography and sense of balance are essential.

Instructional designer: This person plays an important role during the process of developing Computer Based Training (CBT) packages; it is he who decides how a CBT is structured and what kind of learning model should be implemented.

Programmer: A programmer, typically called integrator, is responsible for assembling all the multimedia elements together. He is the person who corresponds the sounds with the buttons, puts the visuals in the right place, gives the transition effects between sequences, etc. In short he puts the entire jigsaw puzzle together.

Quality checker: The person who ensures that the presentation is totally error free and meets the specifications stated. He looks for bugs in the programming, points out errors in the links, and basically ensures an error-free presentation.

Proof reader: His role is similar to that of the Quality Checker. He ensures that the language and grammar used for the presentation is absolutely error free and in the right perspective. He ensures that the language is easy to read, and conveys the right message to the users.

The team members involved in a specific project could include a Creative Director, Graphics Designer, Instructional Designer, Programmer, Proof Reader and Quality Checker (see box, 'The Team').

While the general picture is that the designer and the script writer form the crux of the entire project, a major portion of the responsibility also falls on the programmer. He works in the background, doing the groundwork and needs to understand the complexity of work involved. He ensures that ideas put forward by the team are implemented in right earnest and that the project is executed as it was conceptualised.

Multimedia programming or authoring is essentially writing the scripting or programming code that integrates the various elements of multimedia content, such as text, voice overs, background music, images, animation and video. Precision timing is of the essence here, what with all the fancy graphics, movie clips and sounds that come embedded in the multimedia applications available today.

Good multimedia content is distinguished by the smoothness with which the audio ties in with the video or animation, and the way everything happens without any lag or jerkiness. You would have noticed this on CD-ROMs and Web pages—the moment your cursor moves over a certain point or co-ordinate axis, you see pop up messages, sound applications and voice over enabled, buttons changing colour and the entire interface changing as required to enhance the level of interactivity. The programmer has to take these aspects into consideration and see to it that all the elements work in proper synchronicity.

The learning curve

"Multimedia programming involves lots of creativity and hard work. A programmer should have a sound programming knowledge base in OOPs (Object Oriented Programming Languages), a strong sense of logic with an eye for aesthetics, besides a healthy attitude towards teamwork," says Kiran Nayak, Senior Project Manager, Indiagames.com. A basic knowledge of designing aspects such as colour patterns and designing tools is essential for a programmer as he would have to interact with the designers and almost the whole team.

To be a good multimedia programmer, needless to say, you should have mastery over at least one multimedia authoring system. An authoring system is basically a program which has

pre-programmed elements for the development of interactive multimedia.

Director is one multimedia authoring software that has been around for a long time now and has made life easier for programming greenhorns as well as experts. Director can be customised according to one's requirements. With customisation, it is faster and easier to work with the various commands and templates, normally called libraries. These libraries are very useful as they help combine multiple authoring and tracking views, and project management functions that are associated with such authoring packages.

For any aspiring multimedia programmer, especially a Director programmer, a thorough understanding of Lingo programming is a must.



A multimedia programmer should have sound programming knowledge, a sense of logic and an eye for aesthetics

Kiran Nayak

Senior Project Manager, Indiagames.com

Lingo programming is the scripting language for Director, which helps in providing interactivity to the application. You can even do a certain amount of basic animation using Lingo in Director, though there are specialised software for high-level animations. A Lingo programmer is most sought after for any kind of development, be it game development, CD.

Two other popular multimedia authoring software are Authorware and Toolbook, with Authorware being the more popular of the two. A Macromedia product, this software is an all out logical tool that makes programming for CBT and generating quizzes a cakewalk. Authorware provides you with readymade code that you can drag and drop to control certain events in the package.

You should also know languages that work across platforms. Cross-platform compatibility is an issue with multimedia packages as such packages were originally developed for the Mac but are now used extensively on the PC as well. With the growing effectiveness of the Net as a medium and programs and applications becoming more Web-enabled, there is a growing trend for writing applications that can be viewed and worked upon various platforms such as handheld devices, Pocket PCs and wireless devices.

Flash is another software that is widely used for jazzing up the look and feel of Web sites. Scripting in Flash is still in the primitive stages

